

REGULATORY DOSSIER SUBMISSION

Dossier ID: DOS-DA09F641E0

Country: Tanzania

Submission Date: 2025-08-23

Product: Ceftrax (ceftriaxone) | ATC: J01DD04 | Strength: 1 g/vial

Applicant: Alpha Therapeutics | Manufacturer: Ouaga Public Health Formulations

Policy Label: reject_and_return | Risk Score: 0.92

AMR Stewardship: AWaRe=watch | Unmet Need=moderate | GLASS Trend=rising | Watch Similarity=high

SECTION CONTENT

[1] m1_application_admin - Application Form and Administrative Information

Labels: presence=present; length=length_ok; correctness=correct

Application dossier for Ceftrax (ceftriaxone) submitted on 2025-08-23 to the Tanzania authority. Applicant: Alpha Therapeutics. Proposed dosage form: injectable; strength: 1 g/vial; ATC: J01DD04. Administrative declarations are present but naming conflict remediation documents are incomplete. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. The applicant confirms alignment with applicable technical guidance and regional submission standards. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. Applicant authorization documents and legal attestations were reconciled against the named manufacturing parties. Method validation records and quality controls were reviewed for internal consistency. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. Regional administrative forms were checked for signature completeness and dossier version alignment. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. Adverse events were coded using the Medical Dictionary for R

[1] m1_manufacturer_gmp - Manufacturer and GMP Evidence

Labels: presence=present; length=length_ok; correctness=incorrect

Error tags: gmp_non_compliant

Inspection findings for Ouaga Public Health Formulations identified critical GMP deficiencies. Inspection date: 2022-07-23. Status: non_compliant. CAPA package is incomplete and verification of closure is pending. Method validation records and quality controls were reviewed for internal consistency. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. The GMP evidence package includes certificate status, inspection scope, and manufacturing-site accountability records. Inspection history, CAPA closure records, and site responsibilities were reconciled across the manufacturing chain. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. The applicant confirms alignment with applicable tec

[1] m1_product_information - Product Information and Naming

Labels: presence=present; length=length_ok; correctness=incorrect

Error tags: inn_infringement

Proposed product name is Ceftrax, with INN listed as ceftriaxone. Name similarity review indicates potential confusion with protected and existing nomenclature classes. The submission did not include a sufficient differentiation and risk mitigation plan for look-alike and sound-alike naming concerns. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. The applicant confirms alignment with applicable technical guidance and regional submission standards. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. The risk management plan include

[2] m2_quality_overall_summary - Quality Overall Summary

Labels: presence=present; length=length_ok; correctness=correct

This Quality Overall Summary (QOS) provides a critical evaluation of the chemistry, manufacturing, and controls (CMC) data for ceftriaxone. The control strategy justifies the proposed specifications for both the active substance and the finished product, emphasizing critical quality attributes (CQAs) and critical process parameters (CPPs). Impurity profiles, including degradation products and potential genotoxic impurities, have been thoroughly characterized. Batch analysis data from three commercial-scale validation batches demonstrate consistent manufacturing performance and compliance with all release criteria. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. Method validation records and quality controls were reviewed for internal consistency. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. The applicant confirms alignment with applicable technical guidance and regional submission standards. All referenced documents are available in the dossier annex and traceable by section identifier. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. Stability testing was conducted in climatic zones III and IV, reflect

[2] m2_clinical_overview - Clinical Overview and Benefit-Risk Summary

Labels: presence=present; length=length_ok; correctness=correct

The Clinical Overview synthesizes the efficacy and safety data supporting the use of Ceftnova for the treatment of uncomplicated malaria. The pivotal clinical development program comprised 4 well-controlled studies. Based on the integrated analysis, the reported clinical outcome category is 'endpoint_met'. The overall benefit-risk profile is considered highly favorable for the target patient population. Safety findings were consistent with the known pharmacological class effects, and appropriate risk minimization measures have been integrated into the proposed prescribing information. GLASS-aligned surveillance trend is rising, and cross-resistance risk relative to cefotaxime is assessed as high. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy,

precision, and specificity. No new safety signals were identified during the extended open-label follow-up period. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. Benefit-risk language was reviewed for consistency with the submitted efficacy and safety summaries. The target population demographics are broadly representative of the intended commercial patient cohort. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. The clinical overview links the therapeutic rationale to the pivotal evidence

[3] m3_api_quality - API Quality and Control Strategy

Labels: presence=missing; length=missing; correctness=incorrect

Error tags: missing_critical_section

[MISSING SECTION]

[3] m3_fpp_manufacturing - FPP Manufacturing Process and Controls

Labels: presence=present; length=length_ok; correctness=correct

The finished pharmaceutical product (FPP) manufacturing process utilizes standard, scalable unit operations. A formal quality risk assessment (e.g., FMEA) was conducted to identify critical process parameters (CPPs) that impact critical quality attributes (CQAs). Process performance qualification (PPQ) reports for three consecutive commercial-scale batches verify that the process is maintained in a state of control. In-process controls, intermediate hold-time studies, and packaging validation data are presented to substantiate the robustness and reproducibility of the commercial manufacturing operations. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. The applicant confirms alignment with applicable technical guidance and regional submission standards. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. All referenced documents are available in the dossier annex and traceable by section identifier. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. Method validation records and quality controls

[3] m3_stability - Stability and Shelf-Life Justification

Labels: presence=present; length=length_ok; correctness=correct

A comprehensive stability program has been executed to justify the proposed shelf-life and storage conditions. Data from both accelerated (40°C/75% RH for 6 months) and long-term storage conditions (spanning up to 36 months) are presented for multiple primary stability batches. Statistical evaluation using linear regression and poolability tests confirms that degradation trends remain well within the acceptable specification limits. Photostability and freeze-thaw cycling studies demonstrate that the product does not require specific handling precautions. A post-approval stability protocol commitment is included. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. Risk mitigation activities are documented with ownership, timelines, and follow-up checkpoints. Method validation records and quality controls were reviewed for internal consistency. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. Stability testing

was conducted in climatic zones III and IV, reflecting the intended distribution markets. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. The applicant confirms alignment with applicable technical guidance and regional submission standards. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. All referenced documents are available in the dossier annex and traceable by section identifier. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. Where applicable, protocol deviations were assessed for impact on i

[4] m4_nonclinical_summary - Nonclinical Study Summary

Labels: presence=present; length=length_ok; correctness=correct

The nonclinical testing program evaluated the primary pharmacodynamics, secondary pharmacodynamics, safety pharmacology, and comprehensive toxicology profile of the compound. Repeat-dose toxicity studies in two mammalian species established clear no-observed-adverse-effect levels (NOAELs). Genotoxicity testing (Ames and in vivo micronucleus assays) yielded negative results. Reproductive and developmental toxicity studies revealed no teratogenic signals at clinically relevant exposures. The calculated safety margins support the proposed clinical dosing regimen without major safety concerns. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. The manufacturing site operates under a fully implemented Pharmaceutical Quality System (PQS) aligned with ICH Q10. Stability testing was conducted in climatic zones III and IV, reflecting the intended distribution markets. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. All referenced documents are available in the dossier annex and traceable by section identifier. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. The applicant confirms alignment with applicable technical guidance and regional submission standards. Risk mitigation

[5] m5_trial_listing - Tabular Listing of Clinical Studies

Labels: presence=present; length=length_ok; correctness=correct

The tabular listing of clinical studies provides a comprehensive inventory of all trials conducted in support of the proposed indication (uncomplicated malaria). This includes Phase I pharmacokinetic/pharmacodynamic studies, Phase II dose-finding studies, and the pivotal Phase III efficacy and safety trials. For each study, the table details the protocol identifier, study design, randomization ratio, number of enrolled subjects, study duration, and current completion status. GLASS-aligned surveillance trend is rising, and cross-resistance risk relative to cefotaxime is assessed as high. The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. All referenced documents are available in the dossier annex and traceable by section identifier. Risk mitigation activitie

[5] m5_pivotal_trial_reports - Pivotal Clinical Trial Reports

Labels: presence=present; length=length_ok; correctness=correct

Detailed clinical study reports (CSRs) for the pivotal efficacy trials are provided. These randomized, double-blind, actively-controlled trials evaluated the primary and secondary endpoints in accordance with the pre-specified statistical analysis plan (SAP). The trials resulted in a formal outcome category of 'endpoint_met', demonstrating robust treatment effects in the intent-to-treat (ITT) and per-protocol populations. Subgroup analyses across age, gender, and baseline disease severity strata were consistent with the primary findings. Serious adverse events (SAEs) and discontinuations were systematically analyzed and adjudicated. GLASS-aligned surveillance trend is rising, and cross-resistance risk relative to cefotaxime is assessed as high. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. Adverse events were coded using the Medical Dictionary for Regulatory Activities (MedDRA) version 24.0. The trial reports connect efficacy conclusions to the prespecified analysis framework and observed safety findings. All referenced documents are available in the dossier annex and traceable by section identifier. The pivotal study narratives align endpoint definitions, analysis populations, and outcome interpretation. The primary efficacy endpoint was met, demonstrating a statistically significant improvement over placebo ($p < 0.001$). The container closure system consists of PVC/PVDC-Aluminum blisters, compliant with pharmacopoeial standards. Pharmacokinetic parameters were derived using non-compartmental analysis from the concentration-time profiles. Method validation records and quality controls were reviewed for internal consistency. The analytical procedure was validated in accordance with ICH Q2(R1) guidelines, demonstrating acceptable accuracy, precision, and specificity. The risk management plan includes routine pharmacovigilance and proposed educational materials for healthcare professionals. A randomized, double-blind, parallel-group study design was employed to minimize selection and observer bias. Data were

[5] m5_bioequivalence - Biopharmaceutics and Bioequivalence Evidence

Labels: presence=present; length=length_ok; correctness=correct

Biopharmaceutics data substantiate the formulation development bridging between clinical trial materials and the proposed commercial product. In vivo bioequivalence studies demonstrated that the 90% confidence intervals for the geometric mean ratios of C_{max} and AUC parameters fell entirely within the standard acceptance criteria of 80.00% to 125.00%. Additionally, multi-point comparative dissolution profiles across physiological pH ranges (pH 1.2, 4.5, and 6.8) confirmed similarity ($f_2 > 50$). The bioanalytical methods used for plasma quantification were fully validated. Data were analyzed using a mixed-effects model for repeated measures, adjusting for baseline covariates. Where applicable, protocol deviations were assessed for impact on interpretation of outcomes. Statistical significance was set at an alpha level of 0.05, and all tests were two-sided. Adverse events were coded using the Medical Dictionary f